

Artificial Greenspace and Urban Cooling in Palm Springs, CA

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Introduction

- Palm Springs is located in a desert environment in southern California
- Artificial greenspaces exist in Palm Springs, such as golf courses, which maintain very different environmental standards
- Greenspaces use intensive care through water resources and monetary requirements

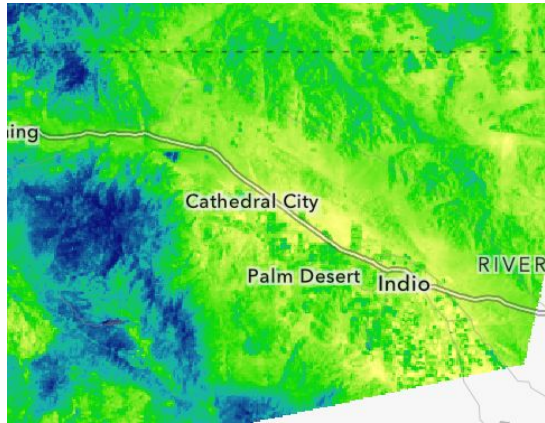
Purpose

- High temperatures and limited water resources play a large role in the daily lives of those in Palm Springs
- Greenspace in Palm Springs requires irrigation for its upkeep
- Are these greenspaces exhibiting significant land surface temperatures lower than their surroundings? Is there any unintentional benefit to adjacent areas?

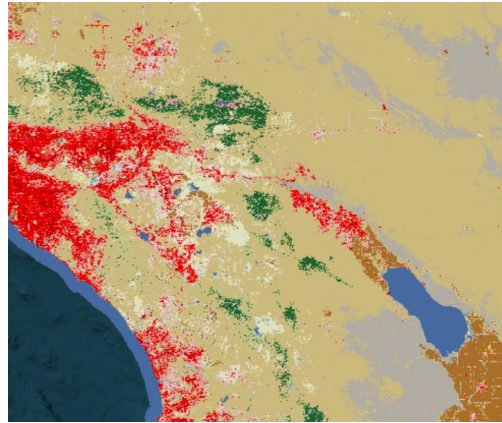


Data

Landsat 8 Thermal Image



NLCD



OpenStreetMap



Methodology

- Calculating average land surface temperatures for golf courses, and for desert areas
 - Using zonal statistics
- Defining adjacent desert areas to golf courses of non-irrigated land
 - Buffering of vector data

Anticipated Outcomes

- A thermal map visualizing the varying land surface temperatures in Palm Springs
 - Expectation of not showing LST that calls for the resource use
- Clear distinction of golf course greenspaces
- Quantified comparison of land surface temperatures of golf courses and their adjacent areas

Sources

<https://www.openstreetmap.org/relation/2866485#map=14/33.82779/-116.53808>

<https://www.mrlc.gov/viewer/>